MILITARY SPECIFICATION

GUN, AIR DEFENSE ARTILLERY, 20MM: TOWED, M167A1

This specification is approved for use by the Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Air Defense Weapon System which consists of an electrically powered, six-barrel, 20 millimeter (mm) gun with linked feed and ammunition system, radar range, lead computing sight and fire control system mounted on a towed, two wheel vehicle. The m167A1 towed, 20MM, Air Defense Artillery Gun is herafter referred to as the weapon system.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form apart of this specification to the extent specified herin.

SPECIFICATIONS

Military	
MIL-W-13855	Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-G-45500/3	Gun, Automatic, 20MM: M168
MIL-I-45607	Inspection Equipment, Acquisition Maintenance and Disposition of
MIL-R-50651	Radar Set, AN/VPS-2: 10549058 (Range Only Radar)
MIL-S-60661	Sight, Lead-Computing, M61 for Gun, Anti-Aircraft Artillery, Self-propelled, 2omm, Xm163 and Towed, 20mm, XM167
MIL-G-60662	Generator, Sight Current
MIL-F-60664	Fire Control for Gun, Anti-Aircraft Artillery, Towed, 20mm, XM167 and Gun, Anti-Aircraft Artillery, Self-propelled, 20mm, XM163

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STANDARDS

Federal		
Fed.Std. No. 356	-	Commercial Packaging of Supplies and Equipment
Military		
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-461.	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	-	Environmental Test Methods

DRAWINGS

Rock Island Arsenal		
F12007503	-	Gun, Air Defense Artillery, Towed, 20mm, M167A1

(Product drawings referenced in this specification form apart of Drawing F12007503.)

PUBLICATIONS

Rock Island Arsenal		
P12007501	-	Packaging Data Sheet for Gun, Air Defense Artillery, Towed, 20MM: M167A1
QE-G-DL12007503		Quality Engineering Data List

(Copies of specifications, standards, drawings and publications required by contractors in connection With specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

3.2 Material, construction and design. The weapon system shall conform to the material, construction and design requirements specified herein, on Drawing F12007503 and in MIL-W-13855.

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3.2.1 Gun, M168. The 20mm, automatic, M168 gun shall be in accordance with MIL-G-45500/3.

3.2.2 Range radar AN/VPS-2. The AN VPS-2 range radar shall be in accordance with MIL-R-50651 and MIL-F-60664.

3.2.3 Lead computing sight, M61. The M61 lead computing sight shall be in accordance with MIL-S-60661 and MIL-F-60664.

3.2.4 Sight current generator. The sight current generator shall be in accordance with MIL-G-60662 and MIL-F-60664.

3.2.5 Running gear. The weapon system running gear wheel nuts shall be torqued in accordance with the applicable drawing requirements. Tire pressure shall be in accordance with drawing requirements.

3.2.6 Rear lights. The rear lights, electrical harness and connector shall be installed in accordance with drawing requirements. There shall be no shorts, and continuity shall be in accordance with drawing requirements.

3.2.7 Hydraulic system. The hydraulic system shall show no evidence of oil leakage at the hydraulic fittings and connections when under operating pressure in accordance with the drawing requirements.

3.2.8 Ammunition feed system. The feed system shall consist of a 500 round ammunition can assembly, feed chute, link chute and a case chute installed in accordance with drawing requirements.

3.3 Performance characteristics.

3.3.1 Warm-up. The weapon system shall require no more than one hour to warm up from -40°F, and no more than four minutes at +60°F or above, to become operable in the test, radar, manual, external and ground mode. Testing shall be as specified in 4.5.2.2.

3.3.1.1 Test mode. The weapon system, when placed in the TEST MODE, shall indicate that the weapon system is operational by the illumination of test completed light. Testing shall be as specified in 4.5.2.2.

3.3.1.2 Radar mode. The weapon system, when placed in the Radar MODE, shall permit the following functions when actuating either or both of the action switches located in the hand grips: Testing shall be as specified in 4.5.2.2.

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a. Slew the gun and gun mount to acquire the target in the reticle of the lead-computing sight.
b. Cause the radar to radiate by depressing and holding the radar-radiate foot switch while continuing to track the target. The ready-to-fire indicator shall be energized when the target is acquired.
c. Fire the gun by depressing the right hand trigger switch.
d. Return the system to the standby condition by releasing the trigger switch, the action switches, and the radar-radiate foot switch.

3.3.1.3 Manual mode. The weapon system, when placed in the MANUAL MODE, shall permit the following functions: Testing shall be as specified in 4.5.2.2.

a. The sight reticle shall have correct directional response to each potentiometer setting.
b. Slew the gun and gun mount to acquire the target in the sight reticle after actuating either or both of the action switches located in the hand grips.
c. Complete the firing circuit by depressing the right hand trigger switch and actuating either or both of the action switches located in the hand grip.

3.3.1.4 External mode. When the weapon system is placed in the EXTERMAL MODE and the remote control assembly installed, the remote operator shall provide the range setting and energize the ready-to-fire Indicator lamp in the sight. Corresponding settings in the external mode sha1l result in the same directional response obtained in the manual mode. Testing shall be as specified in 4.5.2.2.

3.3.1.5 Ground mode. The weapon system, *when* placed in the GROUND MODE of operation, shall not require functioning of the inverter or the radar. The ground mode shall permit the following functions: Testing shall be as specified in 4.5.2.2.

a. The sight to be mechanically caged at a superelevation of -7 +2 milliradians.
b. Illumination of the sight reticle.
c. An operator to slew, elevate, and aim the gun after actuating either or both of the action switches located in the hand grips.
d. An operator to fire the gun by depressing the right hand trigger switch and activating either or both of the action switches located in the hand grips.

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3.3.2 Design functions. The weapon system controls and indicators shall sequentially function by manual and automatic means to accomplish the events in each mode of operation as specified on Figure 1. Functional requirements shall be in accordance with 3.3.2.l through 3.3.2.4 and MIL-F-60664. Testing shall be as specified in 4.5.2.2.

3.3.2.1 Sight caging.

3.3.2.1.1 Mechanical caging of a sight, When the caging knob is positioned in clockwise position the sight shall be caged mechanically to lock the sight gyro-in a position corresponding to O + 2 milliradians in azimuth and -7 + 2 milliradians in elevation with respect to the electrical cage position.

3.3.2.1.2 Electrical caging of sight. The sight shallbe caged electrically in a zero lead angle position, as specified in MIL-S-60661, when activated by depressing the control button in the left hand grip assembly.

3.3.2.1.3 Electromechanical interlock. The sight shall include an electromechanical switch which shall prevent electrical power being. applied to the system while the gyro is in uncaged position. A bypass circuit, using a shorting plug, shall be part of the interlock Circuit.

3.3.2.2 Ready-to-fire indicator. The ready-to-fire indicator, located on the lead-computing sight, shall energize:

a. When all of the following conditions are present in the radar mode:

(1) Radar lock-on signal shall be present
(2) Target range is between 220 meters and 2200 meters incoming, and between 200 meters and l600 meters outgoing
(3) Lead angle shall than 25 degees as defined in MIL-S-60661
(4) The acquisition time delay is completed as defined in MIL-S-60661
(5) Gun elevation angle is less than 80 degrees

b. When turned on by the remote control switch (external mode).

c. At all times in manual mode.

3.3.2.3 Radar operational indicator. The radar operational indicator shall be energized when the system is in the test, radar or manual mode of operation and the radar is warmed up and ready to radiate. The ready-to-radiate status of the radar shall be achieved in not more than two minutes after selection of the radar, test or mnual mode.

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3.3.2.4 Radar radiate foot switch. The radar radiate foot switch shall cause the radar to radiate when depressed and the radar shall return to standby when pressure is released. CAUTION: Permanent injury may occur if subjected to direct radiation within two feet of the radar antenna feed horn. During the radiate period, operators shall not direct the antenna toward any personnel within twenty feet.

3.303 System reaction time. With the following two conditions present, the weapon system shall slew 180 degrees and acquire, track, and fire on the target in not more than 10.5 seconds: Testing shall be as specified in 4.5.2.2.

a. Target velocity and lead angle equivalent to a sight sensitivity, as defined in MIL-S-60661, of less than three seconds.
b. Target within range and of a size that is sufficient for radar lock-on.

3.3.4 System fire control. The weapon system fire control operation shall be in accordance with MIL-F-60664. Testing shall be as specified in 4.5.2.2.

3.3.5 Rates of fire. The weapon system shallpermit two rates of fire which shall be selectable by rotating the FIRING RATE switch on the control palel. The rates of-fire at 70°F ± 30° amibient shall be as follows: Testing shall be as specified in 4.5.2.2.

a. Low rate of fire: 400 to 1300 shots per minute.
b. High rate of fire: 300 to 4000 shots per minute.

3.3.6 Acceleration to firing rate. The weapon system shall attain a rate of 2700 shots per minute within 0.35 seconds from initial gun barrel cluster rotation. Testing shall be as specified in 4.5.2.2.

3.3.7 Burst limiting. The weapon system shall permit preelection for the minimum and maximum rounds to be fired while operating in the high rate of fire. The number of rounds to be fired shall be capable of adjustment and be selected by a high-burst limit position on the firing rate switch, located on the control panel, as follows: Testing shall be as specified in 4.5.2.2.

a. Hi-Burst Limit 10: 8 to 12 rounds.
b. Hi-Burst Limit 30: 27 to 33 rounds.
c. Hi-Burst Limit 60: 54 to 66 rounds.
d. Hi-Burst Limit 100: 90 to 100 rounds.

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3.3.8 Field of fire. The weapon system shall have a field of fire defined for both azimuth and elevation as follows: Testing shall be as specified in 4.5.2.2.

a. Azimuth : 360° in either direction.
b. Elevation: from 5 + 1 degrees below to 80 + 1 degrees above the horizontal reference line of the carriage. The electrical switch assembly in the elevation drive system shall limit powered elevation (and depression) travel from one to three degrees before the mechanical stops contact.

3.3.9 Slewing rate. The weapon system slewing rate shall be at least 75 degrees per second in azimuth in the ground mode. The minimum slewing rate in all other modes shall be 70 degrees per second in azimuth. The slewing and tracking rate shall be 60 degrees per second in elevation. The azimuth velocity shall be measured during the fourth revolution of the gun mount with the gun mount making six revolutions total. Testing shall be as specified in 4.5.2.2.

3.3.10 Gun mount acceleration. With maximum hand control signal, the initial gun mount acceleration shall be at least 160 degrees per second squared (second^2) in azimuth, and 160 degrees per second squared (second^2) in elevation; when measured from the start of gun mount movement. Initial acceleration is defined as the slope of a line through a point tangent to the velocity curve normalized for a 0.2 second increment. Testing shall be as specified in 4.5.2.2.

3.3.11 Rounds counter. The weapon system shall have a total rounds expended counter. The rounds counter shall indicate in 1000-round increments and shall be nonresettable, totaling at 20,000 rounds expended. Testing shall be as specified in 4.5.2.2.

3.3.12 No damage stall and stop. The weapon system shall incur no damage when not firing and operated under the following conditions: Testing shall be as specified in 4.5.2.2.

a. When the gun mount and gun are in positive travel lock position.
b. When the gun is run into the elevation stops at the slewing rate specified in 3.3.9.

3.3.13 Firing interrupter. The firing interrupter shall interrupt firing in predetermined zones of azimuth and elevation when activated by extending the applicable rods in the firing interrupter assembly. Testing shall be as specified in 4.5.2.2.

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3.3.14 Elevation limit switch and stops. Elevation limit switches and mechanical stops shall assure conformance to requirements specified in 3.3.8.b. Testing shall be as specified in 4.5.2.2.

3.3.15 operational capability. The weapon system shall operate at both the high and low firing rates specified in 3.3.5 and acceleration to firing rates specified in 3.3.6 at any ambient temperature from -40°F to +125°F. The permissible degradation of the performance specified in 3.3.5 and 3.3.6 shall be 5, 10 and 15 percent respectively at 0°F, -20°F and -40°F. Testing shall be as specified in 4.5.2.2.

3.3.16 Declutching feeder. The declutching feeder shall delink the linked ammunition coming from the feed chute, feed the rounds into the gun, and guide the empty links into the exit chute. With the GUN CLEAR stitch set in AUTO position, the declutching feeder shall completely disengage the ammunition feed system during the clearing cycle at the end of each burst cycle. Testing shall be as specified in 4.5.2.2.

3.3.17 Ammunition feed system. The feed chute sha1l carry the linked ammunition from the can assembly; the link chute assembly shall carry the empty links from the feeder assembly; and the case chute assembly shall carry the empty cases from the M168 gun. There shall be no jamming or binding throughout operation of the ammunition feed systems Testing shall be as specified in 4.5.2.2.

3.3.18 Azimuth indicator. The azimuth indicator shall freely position-manually and shall retain its manually preset position with respect to the turret during the firing of the weapon system. The lower pointer of the azimuth indicator shall be in zero degrees position when the gun barrel cluster is in travel lock position. The upper two pointers shall move to correspond with the traverse of the gun mount and are resettable. Also, the outer most ring called the "gunner aid dial" is resettable. Testing shall be as specified In 4.5.2.2.

3.3.19 Clearing.

3.3.19.1 Gun clearing switch (Power). When the GUN CLEAR switch is set in the AUTO position, the weapon system shall operate automatically to clear the gun. When the GUN CLEAR switch is held in the OPERATOR position, the gunner shall be capable of clearing the gun by depressing the action switch in the elevation handle. Testing shall be as specified in 4.5.2.2.

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3.3.19.2 Brake-clear and brake switch (Manual). The BRAKE-CLEAR AND BRAKE switch is a three position switch which is spring loaded to the OFF (center position. The BRAKE switch, when held in BRAKE position, shall release the gun drive motor brake and allow rounds to be cycled through the gun by manually rotating the gun barrel cluster. The BRAKE-CLEAR switch, when held in the BRAKE-CLEAR position, shall release the gun drive motor and energize the declutching feeder solenoid which shall prevent rounds from being fed into the gun. The spring loaded OFF position (center position) shall prevent the gun barrel cluster from being manually rotated. Testing shall be as specified in 4.5.2.2.

3.3.20 Safety travel lock. The positive safety travel lock when engaged shall prevent unintentional slewing of the mount and barrel assembly rotation. Testing shall be as specified in 4.5.2.2.

3.3.21 Remote safe switch. A remote safe switch (when attached) shall prevent unintentional firing of the gun by replacing the firing circuit arming connector with a manually controlled switch. This switch shall permit remote control of the ability to fire the gun. Testing shall be as specified in 4.5.2.2.

3.3.22 Sustained use. When operated within an ambient temperature of 32 to 125°F with battery power only and the firing rate switch in the high position, the weapon system shall fire three 60 round bursts at 7 minute intervals. The ambient temperature at which these firings are conducted shall be within 10°F of the ambient temperature at which rates of fire were established in 3.3.5. The system battery power shall remain ON during the intervals between bursts. Degradation of the third burst shall not exceed 15 percent of the performance as recorded for the system in meeting 3.3.5, 3.3.6 and 3.3.7. Testing Shall be as specified in 4.5.2.2.

3.3.23 Low voltage warning indicator and interrupting circuit. The low voltage warning indicator shall be energized during periods of low system voltage. When the voltage drops below 22 vdc minus l 22 vdc, the indicator light shall flash. When the voltage drops below 20.5 vdc ± .21 vdc the indicator light shall be constantly illuminated and shall interrupt power to the sight current generator, radar, inverter, and sight gyro. Testing shall be as specified in 4.5.2.2.

3.3.24 Low voltage test/override switch. The low voltage test/override switch shall perform as indicated in the following positions: Testing shall be as specified in 4.5.2.2.

a. NORMAL position: The switch shall permit the actions noted in 3.3.23.

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b. OVERRIDE position: The interruped normal circuit shall be by-passed and the available system power shall be restored to the sight current generator and the radar inverter when the following sequence is performed:

(1) System switch turned from ON to OFF to ON;
(2) Low voltage switch is moved to override position while warning light is flashing.

c. TEST position: A simulated low voltage circuit shall. be activated. The warning indicator light shall flash for approximately four seconds then become constantly illuminated.

3.3.25 Tracking rate. The weapon system shall be capable of tracking targets with an angular velocity of velocity-range (V/R) greater than on. Testing shall be as specified in 4.5.2.2.

a. The weapon system shall move smoothly during tracking operations that will require gun movements in either elevation, azimuth or a combination of each at rates as low as 0.1 degrees per second.
b. The gun drift in any vertical direction shall be no greater than 5.0 milliradians during a 100 round burst while slewing in azimuth at a rate of 5.0 ± 0.1 degrees per second.

3.3.26 Boresighting. The weapon system shall be boresighted by utilizing a distant aiming point of at least 2500 meters. Testing shall be as specified in 4.5.2.2.

3.3.27 Hand brakes. The *weapon* system hand brakes, when engaged, shall prevent the fully equipped and serviced weapon system from rolling *on a 60* percent slope of brushed concrete. The brake-shoes shall not drag when the brake lever is in the OFF position. Testing shall be as specified in 4.5.2.2.

3.3.28 Towing. The weapon system, prepared for travel, shall withstand four 1000 mile test cycles under conditions specified in Table I without damage or loosening of parts. The weapon system shall be capable of ascending and descending slopes of 60 percent and negotiating side slopes of 20 percent. The weapon system shall be capable of traversing ditches where the sum of the slopes forming the ditch does not exceed 60 percent and the straight away of each slope is at least as long as the combined length of the weapon system and mover. Testing shall be as specified in 4.4.

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TABLE I

Road Type	Speed (MPH)	Distance (Miles)
Gravel	20 - 30	300
Belgian Block	15 - 20	50
Hilly Cross-country	5-15	200
Level Cross-country	10-20	250
Paved	45 Maximum	200
	Total	1000

3.3.29 Fording function. The weapon system shall be capable of fording a depth of 30 inches with leakage into the carriage not to exceed 473 cubic centimeters (one pint) per hour. All components and assembles below the 30 inch fording depth shall not suffer degradation in performance from water. Testing shall be as specified in 4.4.

3.4 Environmental. The weapon system shall meet the performance requirements of this specification as specified in 3.3 during exposure to or after exposure to, environmental conditions specified In 3.4.1 through 3.4.9. Testing shall be as specified in 4.4.

3.4.1 Altitude. The weapon system shall operate without failure at altitudes from sea level to 10,000 feet and shall be transportable at altitudes up to 40,000 feet.

3.4.2 Temperature. The weapon system shall operate without failure in temperatures ranging from 40° to +125°F. The weapon system shall function as specified herein after storage in a temperature of +155°F for a minimum of four hours.

3.4.3 Rain. The weapon system, with no protective cover, shall show no degrdation of performance or evidence of corrosion which would prevent the weapon system from performing its intended function when exposed to rain and wind conditions specified in Procedure I, Method 506 of MIL-STD-810.

3.4.4 Humidity. The weapon system shall show no degradation of performance when:

a. Exposed to relative humidity of five percent at 125°F to 100 percent at 85°F.
b. Stored in a relative humidity of 100 percent at 85°F for a minimum of 48 hours.

3.4.5 Fungus. The weapon system shall be resistant to microorganisms specified in Method 508 of MIL-STD-810.

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3.4.6 Salt fog, The weapon systems shall show no degradation of performance or evidence of corrosion which would prevent the weapon system from performing its intended function when exposed to the effects of a salt atmosphere as defined herein.

a. Temperature, 95°F.
b. Atomization of approximately 3 quarts of salt solution per 10 cubic feet of test chamber volume per 24 hours.
c. The weapon system shall be exposed to the salt fog atmosphere for a period of 48 hours.

3.4.7 Dust. The weapon system shall operate while being subjected to blowing dust with particle size of 0.0001 to 0.0100 mm at wind speeds of 15 knots.

3.4.8 Shock. The weapon system shall withstand, without damage, shock forces encountered during installation and service.

3.4.9 Sand. The weapon systems shall operate while being subjected to blowing sand of particle size from 0.01 to 1.00 mm at a wind speed of 35 knots.

3.5 Interchangeability. Unless otherwise specified on the drawings all parts of the same part number shall be interchangeable. Testing shall be as specified in 4.5.2.3.

3.6 Electromagnetic interference characteristics. The weapon system shall meet the applicable electromagnetic interference characteristics requirements of MIL-STD-461 as specified in the contract (see 6.2). Testing shall be as specified in 4.4.

3.6.1 System and radar compatibility. The gun mount (turret) drive electrical noise generated during traversing and stopping the gun mount shall not produce a false radar “lock-on”. (The ready-to-fire light shall remain off.)

3.6.1.1 The gun mount (turret) drive electrical noise with the mount traversing drive motors locked and the systems in radar mode, the decibel (DB) difference while operating before ambient DB, and after revolving hand controls with action switch depressed maximum right and and maximum left position, shall not exceed 4 DB. Distribution box switch shall be in “test” position. The electrical noise shall not produce a false radar “lock-on”.

3.7 Marking. The weapon system shall be marked in accordance with MIL-W-13855. Each weapon shall be identified by a serial number assigned by the procuring activity (see 6.2).

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3.8 Workmanship. The workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilitv for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4. 5).

4.4 First article inspection, The first article (initial production unit(s) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the Quality Conformance inspection (see 4.5), Towing and Fording test (see 4.6,30), Environmental test (see 4.6.31), Electromagnetic interference test (see 4.6.33) and such other inspection as necessary to determine that ail requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Examination.

4.5.1.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The examination of these parts shall. be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.1.2 Material and construction examination. Each weapon system shall be visually examined to determine compliance with 3.2, utilizing Government approved inspection procedures.

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4.5.1.3 System components examination. Each weapon system shall be visually examined to determine compliance with 3.2.1 through 3.2.8. For Government furnished components, contractor responsibility shall be as prescribed in 4.5.4.

4.5.1.4 Marking examination. Each weapon system shall be visually examined to determine compliance with 3.7.

4.5.1.5 Workmanship examination. Each weapon system shall be examined for proper cleaning, protective coatings, and to determine general quality, completeness of manufacture, assembly and workmanship in compliance with 3.8.

4.5.2 Testing.

4.5.2.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete weapon system exclusive of Government furnished property. If test requirements cited herein are not met, acceptance of the weapon system shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine weapon systems, partially assembled weapon systems and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned weapon system for retest.

4.5.2.2 System testing. Unless otherwise specified (see 6.2), each weapon system shall be subjected to the test methods specified in 4.6.1 through 4.6.29, except only one weapon system randomly selected from each 20 systems produced shall be subjected to the -40°F warm-up temperature (see 3.3.1) using the test method specified in 4.6.1 Failure of any system to meet the requirements of 3.3.1 through 3.3.27 shall be cause for rejection.

4.5.2.3 Interchangeability testing. A sample of three weapon systems selected at random by the Government representative from each 20 systems produced shall be tested for interchangeability (see 3.5),

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Using the test method specified in 4.6.32. After completion of the test and the weapon systems are reassembled, each system shall be subjetted to the testing specified in 4.5.2.2. Failure of any reassembled system to meet the requirements (see 3.5) shall be cause for rejections

4.5.3 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified in the contract (see 5.2), responsibility for acquisition, calibration, maintenance and disposition Of acceptance inspection and test equipment prescribed on QE-G-DL12007503, and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used it must be capable of repetitive measurements to an accuracy of 10% of the total tolerance of the characteristics being inspected.

4.6 Test methods.

4.6.1 Warm-up test. The weapon system shall warm-up to permit operation in the test, radar, manual, external and ground modes to determine compliance with 3.3.1.

4.6.2 Test mode test. The weapon system shall be operated in the test mode. A simulated problem shall be presented to the sight current generator and correctly solved, as indicated by the lighting of the GOOD TEST light to determine compliance with 3.3.1.1.

4.6.3 Radar mode test. The weapon system shall be operated in the radar mode. The rdar and lead computing feature of the sight shall be operating; the radar shall be supplying range data to the weapon system; and the gun mount shall rotate and track the target. The READY-TO-FIRE indicator lamp (located on the sight) shall light, to determine compliance with 3.3.1.2. After release of the trigger switch, action switch and the radar radiate foot switch, the weapon system shall return to the standby condition.

4.6.4 Manual mode test. The weapon system shall be operated in the manual mode. The RANGE and TARGET SPEED controls shall be set at estimated values. The sight gyro shall be electrically caged in accordance with MIL-s-60661 by depressing the sight cage button during turret slew. When the target is acquired in the sight reticle

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the sight cage button shall be released, but tracking shall continue until the firing engagement is complete. The READY-TO-FIRE indicator lamp shall remain lit, to determine compliance with 3.3.1.3 and 3.3.2.1.2.

4.6.5 External mode test. The weapon system shall be operated in the external mode. The weapon system shall receive and use range data supplied by the External Range Control. The READY-TO-FIRE indicator lamp shall light when the external READY-TO-FIRE button is depressed to determine compliance with 3.3.1.4.

4.6.6 Ground mode test. The weapon system shall be operated in the ground mode. The sight shall be mechanically caged 7 mils super-elevation in accordance with MIL-S-60661 and the reticle lamp on the sight shall light to determine compliance with 3.3.1.5 and 3.3.2.1.1.

4.6.7 Design function test. The weapon system shall be cycled through each mode of operation, both manually and automatic, as specified on Figure 1 to determine compliance with 3.3.2.

4.6.7.1 Sight caging test. The sight shall be caged to determine compliance with 3.3.2.1 through 3.3.2.1.3.

4.6.7.2 Sight caging interlock test. The weapon system shall be-tested for sight caging interlock and by-pass circuit to determine compliance with 3.3.2.1.3.

4.6.7.3 Ready-to-fire indicator test. The ready-to-fire indicator shall be tested in the radar, external and manual modes to determine compliance with 3.3.2.2.

4,6.7.4 Radar operational indicator test. The indicator still be tested to determine compliance with 3.3.2.3.

4.6.7.5 Radar radiate foot switch test. The foot switch shall be tested to determine compliance with 3.3.2.4.

4.6.8 Reaction time test. When subjected to the conditions of 3.3.3, the time required for the weapon system to slew 180°, acquire, track and open fire on the target, using dummy ammo, shall be measured and shall not exceed 10.5 seconds, under the stipulation that operating personnel have demonstratable proficiency in the weapon system, to determine compliance with 3.3.3.

4.6.9 System fire control test. The system fire control operation test shall be as specified in MIL-F-60664, to determine compliance with 3.3.4.

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4.6.10 Firing rate test. The FIRING RATE switch on the control panel shall be rotated to the HIGH or LOW rate of fire and the weapon system fired. The rates of fire shall be tested for compliance with 3.3.5 and 3.3.6. Interpolation of rate requirements for systems tested below 40°F is acceptable.

4.6.11 Burst limiting test. The weapon system shall be fired at the high rate in four burst limits to determine compliance 3.3.7.

4.6.12 Field of fire test. The mount and gun shall be operated through all ranges of elevation and azimuth to determine compliance with 3.3.8 The elevation electrical limit switches shall prevent the weapon system from being driven into the mechanical stops.

4.6.13 Slewing rate and acceleration test. The mount and gun shall be slewed in all operational modes. The slew rate and accereration shall be as specified in 3.3.9 and 3.3.10.

4.6.14 Rounds counters test. The rounds counters shall be tested for accurate counting of rounds as specified in 3.3.11. The weapon system shall be fired in bursts as specified in 3.3.7a through 3.3.7d.

4.6.15 Stall and stop test. The weapon system shall be operated through ranges of elevation and azimuth rotation to determine compliance with 3.3.8) 3.3.12 and 3.3.14. The gun mount shall be driven into the limit switch and mechanical stops at maximum slew rate. The mount shall not be damaged.

4.6.16 Firing interrupter test. The firing interrupter assembly (see 3.3.13) shall be visually and manually inspected for secure and correct assembly in the system. Using Government approved test equipment and by simulated firing of the system, automatic interruption of firing shall be cht'eked for each 20° segment of azimuth and each 5° increment of elevation within each segment of azimuth. To assure maximum safety for simulated firing, set controls for interrupted firing and ground mode. Set rods for alternately firing in 20° segments and rotate turret through azimuth for each 5° increment of elevation. Observe that the firing voltage test lamp is "off" or "on" during alternate 20° segments corresponding to the rod settings in the firing interrupter assembly. Failure of the firing interrupter to follow the rod settings shall be cause for readjustment of the interrupter assembly and retest.

4.6.17 Operational capability test. The weapon system shall be operated as described in 3.3.1, 3.3.5 and 3.3.6 to determine compliance with the operational capability of the weapon system as specified in 3.3.15.

4.6.18 Declutching feeder test. The weapon system shall be fired. The ammunition shall be fed to the cannon in compliance with 3.3.16.

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4.6.I.9 Ammunition feed system test. During the tests of 4.6.18, the feed chute; link chute; feeder assembly, and case chute components and system operation shall be inspected for smooth transition of ammunition in compliance with 3.3.16 and 3.3.17.

4.6.20 Azimuth indicator test. The azimuth indicator assembly shall be visually and manually inspected for secure and correct mounting, for free rotation of the dial from O through 64 in a clockwise direction and free rotation in a counter-clockwise direction to determine compliance with 3.3.18. Using system power and hand controls, position the turret to a definite deference point and set the azimuth indicator to 64 on the dial. Slew turret and position at 90° from reference. Continue rotation and reposition on reference point. Compare turret position and indicator readings for accuracy of one half sector on the dial. Repeat the test by rotating turret in opposite direction.

4.6.21 Clearing test.

4.6.21.1 Gun clear switch test. The weapon system shall be fired. The gun shall automatically clear when the GUN CLEAR switch on the control assembly is placed in the AUTO position. When the GUN CLEAR switch is placed in the OPERATOR position, the gun shall clear by pressing. the action switch and actuating the feeder declutching solenoid to determine compliance with 3.3.19.1.

4.6.21.2 Brake-clear and brake switch test. The weapon system shall be checked for proper operation with the BRAKE-CLEAR AND BRAKE switch in each position to determine compliance with 3.3.19.2. Dummy cartridges shall be used to check for proper gun and declutching feeder operation while the power system is on. Placing the switch in BRAKE position shall allow the gun drive motor to release and dummy rounds to be cycled through the gun by manually rotating the gun barrel cluster. Placing the switch in the BRAKE-CLEAR position shall allow the gun drive motor brake to release and energize the declutching feeder solenoid, preventing the rounds from being fed through the gun. When the switch is in center OFF position, the brake shall prevent the gun barrel cluster from being manually rotated.

4.6.22 Safety travel lock test. The mount shall. be tested for compliance with 3.3.20.

4.6.23 Remote safety switch test. The arming connector shall be disconnected at the distribution box and an attempt shall be made to fire the gun. The gin shall not fire. The remote safety switch shall be connected to the distribution box and an attempt shall be made to fire the gun. The gun shall not fire with the remote safety switch open, but shall fire with it closed in compliance with 3.3.21.

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4.6.24 Sustained use test. The weapon system shall be tested for compliance with 3.3.22. Degradation in excess of the specified, or other indications of excessive power loss in the weapon system, shall be cause for rejection.

4.6.25 Low voltage warning indicator and interrupting circuit test. The weapon system shall be tested to determine compliance with 3.3.23.

4.6.26 Low voltage test/override switch test. The weapon system shall be tested to determine compliance with 3.3.24.

4.6.27 Tracking rate test. The weapon system shall be tested to determine compliance with 3.3.25.

4.6.28 Boresighting test. The weapon system shall be subjected to the following boresighting procedure to determine compliance with 3.3.26:

a. Check that the GUN POWER switch is OFF.
b. Set the SYSTEM POWER switch to the ON position and set MODE switch to the RADAR position.
c. Check that the gun is clear. To clear gun, position the BRAKE-CLEAR AND BRAKE switch in the CLEAR AND BRAKE position and rotate the barrels two complete revolutions. Remove any rounds in the case chute.
d. Time the gun in the No. 1 firing position. Hold the BRAKE-CLEAR AND BRAKE switch in the CLEAR AND BRAKE position, depress the gun index pin, and locate the No. 1 barrel (marked on the muzzle clamp) in the firing position which is at the 4 o'clock position when viewed from the muzzle end. Release the BRAKE-CLEAR AND BRAKE switch and index pin. Check that the index pin has returned to normal.
e. Unstow the radar antenna, if stowed. Set MAINT switch to OFF, set the MODE switch to NORMAL, and depress either action switch until the antenna drives out of stow and is positioned on the gun axis. Set MAINT switch to ON.
f. If the radar antenna was unstowed, check to see that the MODE switch is set to NORMAL and the MAINT switch is ON.
g. Select a distant aiming point (at least 2500 meters away) and depress an action switch and train the gun to the approximate position for boresighting.
h. Pull out the shift pin handle on the drum drive assembly and shift to N position. This places the drum drive in neutral. It also interrupts electrical circuits to prevent the elevation, traversing and firing of the gun.

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i. Insert the muzzle adapter from the boresight kit into the muzzle of the No. 1 barrel.
j. Install the boresight telescope in the mandrel assembly. and secure with socket head screws on the mandrel assembly.
k. Insert the mandrel assembly into the muzzle adapter with the eyepiece of the telescope in the horizontal position and press to tighten the adapter and mandrel assembly in the barrel.
l. Release the two azimuth drive motor brakes and the elevation drive rotor brake by positioning the brake release levers perpendicular to the motor axis.
m. Sight through the boresight telescope on a distant point. Rotate the mandrel assembly and boresight telescope 90 degrees to the horizontal position on the opposite side of the barrel. Sight through the boresight telescope to determine that the point sighted initially is still at the cross hairs. If the point is not on the cross hairs, the boresight kit is defective, improperly assembled or improperly mounted.
n. Manually position the gun so the cross hairs of the boresight telescope are level with and 3.5 mils to the right of the distant aiming point.
o. Lock the azimuth and elevation drive motor brakes and check that the gun is still properly aimed.
p. Manually uncage the sight.
q. Adjust the LAMP control assembly for a clear reticle in the sight.
r. Remove only the front and right caps on top of the M61 sight.
s. Adjust the socket head bolts on top of the sight to center the inner reticle of the M61 sight on the distant aiming point. The front adjustment is for elevation and the right adjustment is for azimuth. Move the eye up and down and right and left to check for parallax. If excessive parallax is experienced, refer the sight to organizational maintenance. Note: A second crew member must check through the boresight telescope while the adjustments are being made to assure that the gun does not move.
t. Install the caps on the sight. Note: When boresighting the M134 sight, it is necessary to release the clamping screws in front and rear end of left wall of the M164 Mount.
u. View through the M134 sight and adjust the AZ boresight adjusting screw to bring the cross of the reticle into horizontal alignment with the distant aiming point.
v. View through the M134 sight and adjust the EL boresight adjusting screw to bring the cross of the reticle into verticle alignment with the distant aiming point.

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w. Tighten clamping screws.
x. Remove the boresight telescope, mandrel assembly, and install the boresight telescope in the mount provided in the antenna. Remove muzzle adapter from No. 1 barrel.
y. Loosen the elevation coupling clamp screw which is located on the gun elevation input shaft coupling to the radar antenna. Rotate the radar input shaft to move the horizontal cross hair of the telescope reticle on the distant aiming point. Tighten the screw. Note: A second crew member must check that the sight remains centered on the distant aiming point.
z. Remove the access cover from the traverse boresight adjustment which is located on the reflector and feed housing assembly. Loosen the lock and turn the adjustment screw until the vertical cross hair on the telescope reticle intersects the distant aiming point; then tighten the lock. Note: A second crew member must check that the sight remains centered on the distant aiming point.

aa. Repeat steps (y) and (z) if necesssry to center the reticle cross hairs.
bb . At the antenna stow control, place the MODE toggle switch to STOW.
cc l Using the TRAV toggle switch, drive the antenna to the left approximately 10 degrees.
dd . Place the MODE toggle switch to NORMAL.
ee. At the boresight telescope mounted on the antenna, observe that the cross hairs within the reticle are positioned on the distant aiming point.
ff l Repeat steps (bb) through (ee) except in step (cc) drive the antenna to the right.
gg. At the stow control, place the MODE toggle switch to STOW.
hh. Using the ELEV toggle switch,drive the antenna up approximately 10 degrees.
ii. Place the MODE toggle switch to NORMAL.
jj. At the boresight telescope mounted on the antena, observe that the cross hairs within the reticle are positioned on the distant aiming point.
kk. Repeat steps (gg) through (jj) except in step (hh), drive antenna down approximately 10 degrees.
ll. Replace the access cover on the traverse boresight adjustment and the gun elevation input shaft protective cover.

mm. On the antenna stow control, set the MAINT switch to OFF.
nn. Remove the boresight telescope from the antenna reflector. Caution: Assure that the sun is not shining directly into the AN/VPS-2B sight. Do not remove the AN/VPS lens cover during daylight hours.

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oo. Open the small aperture of the AN/VPS-2B lens cover.
pp. View through the AN/VPS-2B night sight and adjust the azimuth and elevation boresight adjusting screws of the AN/VPS-2B until the distant aiming point is centered on the top center dot of the AN/VPS-2B sight reticle.
qq. Cage the M61 sight.
rr. Set the SYSTEM POWER switch to the OFF position.
ss. On the drum drive assembly, pull out the shift pin handle and place the drum drive in the fire position (shift to F position).
tt. Recheck that there is nothing in the muzzle.

4.6.29 Hand brake test. The weapon system shall be subjected to the hand brake test to determine compliance with 3.3.27, unless documentary evidence is supplied as proof that the brakes were tested during production by an equivalent static torque test with the brake assembly mounted to the running gear.

4.6.30 Towing and fording test. The weapon system shall be towed over terrain and road conditions at speeds specified in 3.3.28 using an M561, M705 or M715 2 1/2 ton truck and coupler adapter (8438335) installed on the carriage. The weapon system shall be tested under the conditions of, and shall comply with the requirements for fording of 3.3.29.

4.6.31 Environmental test. The weapon system shall be tested as specified in 4.6.31.1 through 4.6.31.9 to determine compliance with 3.4.1 through 3.4.9. Where advantageous, naturally occurring environments may be used in lieu of laboratory produced conditions. There shall be no evidence of corrosion, degradation of performance, or damage that would prevent the weapon system from performing its intended function.

4.6.31.1 Altitude test. The weapon system shall be tested to determine air transportability to 40,000 feet and operability to 10,000 feet in accordance with Procedure I, Method 500 of MIL-STD-810.

4.6.31.2 Temperature tests.

4.6.31.2.1 High temperature test. The weapon system shall be tested when stabilized at 1250 after four hours in a storage temperature at 155° in accordance with Procedure 1, Method 501 of MIL-STD-810. The weapon system shall be tested for operation at 120°F in accordance with Procedure II, Method 501 of MIL-STD-810.

4.6.31.2.2 Low temperature test. The weapon system Shall be tested for a low temperature storage condition of -40°F for 12 hours without solar radiation in accordance with Procedure I, Method 502 of MIL-STD-810. At 0°, -20° and -40°F, the degradation in performance shall be not more than 5, 10 and 15 percent, respectively.

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4.6.31.3 Rain test. The weapon system shall be exposed to rain and wind conditions in accordance with prodedure I, Method 505, of MIL-STD-810.

4.6.31.4 Humidity test. The weapon system shall be sujected to humidity conditions for seven cycles of 12 hours each in accordance with Procedure I, Method 507 of MIL-STD-810. The weapon system shall withstand storage at 100 percent relative humidity at a temperature of 85°F.

4.6.31.5 Fungus test. The Weapon system shall be subjected to fungus testing in accordance with Procedure I, Method 508 of MIL-STD 810, unless documentary evidence is supplied as proof that only non-nutrient materials were used or that all parts were suitably treated to resist micro-organisms.

4.6.31.6 Salt fog test. The weapon system shall be subjected to a salt fog test in accordance with Procedure I, Method 509 of MIL-STD-810

4.6.31.7 Dust test. The weapon system shall be subjected to blowing dust with particle size from 0.0001 to 0.0100 mm and wind speeds of 15 knots, in accordance with Procedure I, Method 510 of MIL-STD-810.

4.6.31.8 Shock test. The weapon system shall be shok tested in accordance with Figure 516-1 of Procedure I, Basic Design Test for Ground Equipment, Shock parameters b and d Method 516 of MIL-STD-810.

4.6.31.9 Sand test. The weapon system shall be operated at speeds up to 30 miles per hour in sandy terrain with sind speeds of 35 knots. The weapon system shall be operated, with the vehicle both stationary and moving, in blowing sand having a particle size from 0.01 to 1.00 mm, and with wind speeds of 15 knots or greater. Laboratory produced conditions shall be permitted for this test.

4.6.32 Interchangeability test (see 3.5).

4.6.32.1 Weapon system. Interchangeabliity test prodedures shall provide for three weapon systems, from each twenty systems produced, to be disassembled and reassembled in a prearranged pattern from the parts of each weapon system divided into three groups of nonmating parts distributed in three different trays each containing a complete weapon system. The exchange of parts and assemblies shall result in the maximum exchange in mating parts. The interchangeability test shall include major subassemblies and reparir parts for major subassemblies as specified in Table II. The weapon system shall be functionally tested after reassembly. Any failure of the interchargeability test shall be cause for rejection of the represented lot. Upon authorization form HQ.U.S. ARMCOM. ATTN: DRSAR-QA. Rock Island, IL 61201, an interchangeability retest may be allowed without reconditioning of the lot of weapon systems.

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TABLE II. INTERCHANGEABLE ITEMS

Equipment	Part Number
Support sight assembly	8436972
Equilibrator assembly (2)	12007324
Brake and clear switch assembly	8437323
Drive assembly, elevation	8436847
Drive assembly, azimuth	8437053
Azimuth indicator assembly	8438753
Recoil adapter assembly (2)	8336802
Muzzle clamp assembly	8436804
Declutching feeder assembly	8438645
Servo amplifier assembly (Az. Master)	12000006
Distribution box assembly	8438325
Sight *current* generator	12000239
Slip ring assembly	8438355
Running gear assembly	8438669l

4.6.32.2 Repair parts. Two component parts from each inspection lot of repair parts that are produced specifically for this system shall be-tested for interchangeability in accordance with a Contractor submitted, Government reviewed plan. This plan shall be submitted to the Government no later than 10 calender days prior to scheduled delivery of initial lot of repair parts. Any failure to demonstrate interchangeability shall cause rejection of the lot of parts. No hand refinements will be allowed for retest. Four parts shall be selected from the reconditioned lots for interchangeability testing.

4.6.33 Electromagnetic interference characteristics test. The weapon system shall be tested in accordance with the test methods of MIL-STD-462 to determine compliance with 3.6.

4.6.33.1 System and radar compatibility test. With the system in the "RADAR MODE" the gun mount shall be traversed at maximum slew rate in both clockwise and counterwise direction. (The ready to fire light shall remain off to determine compliance with 3.6.1.) The gun mount electrical noise (radiated emission) shall be measured in an ambient condition, and during traversing the gun mount. The reaiated emission measurement shall be taken in accordance with MIL-STD-462, Method RE05 at the broad band frequencies of 150KHz to 1 GHz. The decible difference shall not exceed four DB in accordance with the requirements of 3.6.1.1:

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a. If the DB increase is less than 4 DB while traversing to the maximum limits and the radar locks-on, the radar sensitivity shall be readusted. This adjustment shall be performed only by a qualified technical representitive of the Government and shall not exceed 2 DB.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of a complete weapon system preserved in accordance with Packaging Data Sheet P12007501 for the level of protection specified in the contract (see 6.2), packed Level B and forwarded as specified in 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be In accordance with packaging Data Sheet p12007501 for the level of protection specified in the contract (see 6.2).

5.3 Commercial packaging. Reservation, packing and marking shall be in accordance with Fed. Std. No. 356.

5.4 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other packaging requirements, as specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M167A1 towed gun, 20mm air-defense artillery is intended for, but not limited to, close range anti-aircraft cover or strafing operation.

6.2 Ordering data. Procurement documents should specify the following :

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the weapon system showing applicable revision dates.
c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Applicable electromagnetic interference characteristics requirements (see 3.6).
e. Block of serial numbers for quantity of weapons systems on order (see 3.7).
f. Quality conformance inspection, if other than as specified (see 4.5).
g. Examination criteria for component and concurrent repair parts (see 4.5.1.1).

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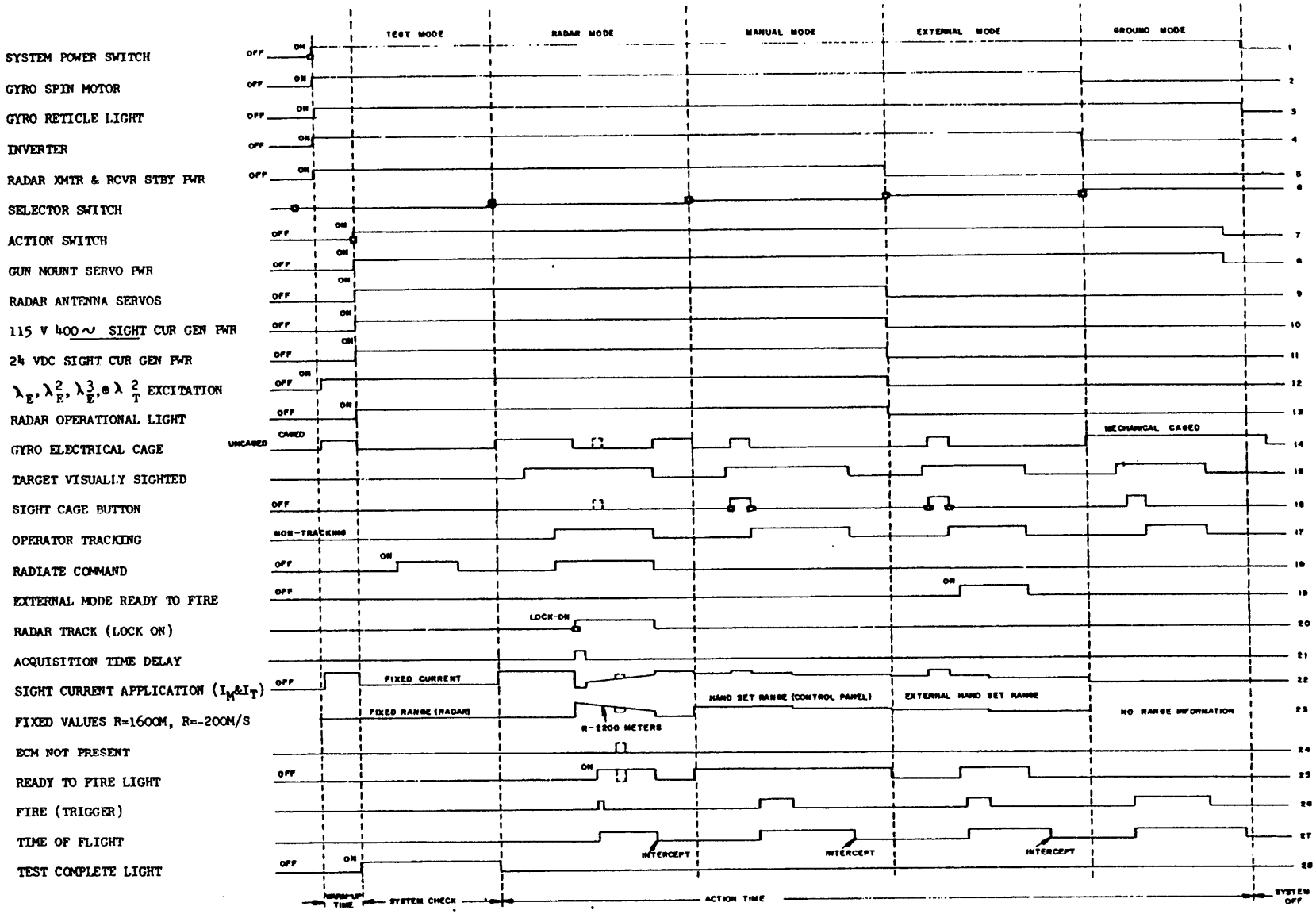
h. Weapon system testing, indifferent (see 4.5.2.2).
i. Packaging examination and testing, if different (see 4.5.3 l
j. Responsibility for acquisition, maintenance, and disposition of inspection equipment, if other than as specified (see 4.5.4.1).
k. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).
l. Packaging instructions for repair parts (see 5.4).

Custodian:
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LEGEND

FIGURE 1. SEQUENCE OF EVENTS DIAGRAM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-G-45981(WC); Gun, Air Defense Artillery, 20MM: Towed, M167A1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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